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Beyond Optical PPG: mmWave Radar Arterial Pulse Sensing

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Abstract

Cardiovascular diseases (CVDs) remain the leading cause of death worldwide, accounting for nearly 20 million deaths in 2022 alone – approximately 32% of all global mortality. Continuous, non-invasive monitoring of vital signs such as heart rate (HR), heart rate variability (HRV), and blood pressure (BP) is therefore critical for early detection and management of CVDs. Optical photoplethysmography (PPG), the sensing modality used in most consumer wearables, is highly susceptible to motion artifacts, ambient light interference, sweat, varying skin pigmentation, and subcutaneous fat thickness, all of which degrade signal quality and downstream physiological estimates.

This feasibility report investigates millimeter-wave (mmWave) radar as a complementary and potentially more robust alternative to optical PPG for wrist-worn arterial pulse sensing. While prior radar-based physiological monitoring systems have shown promising accuracy, they are predominantly evaluated under static conditions using fixed radar configurations that do not adapt to real-time signal quality degradation caused by motion [1–3].

We propose the design and development of a wearable, multimodal sensing prototype that integrates an FMCW mmWave radar (BGT60TR13C), a clinical-grade optical PPG front-end (MAX86176), and an inertial measurement unit (IMU) on a single wrist-worn platform. The proposed system is a closed-loop adaptive radar controller that retunes sensing and processing parameters every coherent processing interval (CPI) using a contextual-bandit algorithm driven by a cardiac signal-to-interference-plus-noise ratio (CSINR) reward and real-time motion state, rather than a fixed processing pipeline. A unified signal-processing and lightweight machine-learning pipeline will be developed to jointly estimate HR, HRV, and BP from the extracted radial-artery pulse waveform.

The system will be validated against clinical-grade reference devices – a Polar H10 chest strap for HR/HRV ground truth and an Omron BP4350 wrist-cuff for BP ground truth – across sitting, walking, and running activity conditions, and across a range of skin tones and subcutaneous fat thicknesses, in order to directly benchmark radar-based sensing robustness against optical PPG.

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Chapter 1

Introduction

1.1 Motivation

Cardiovascular diseases (CVDs) are the leading cause of death worldwide. Nearly 20 million people died from CVDs in 2022 alone, accounting for approximately 32% of all global deaths. Continuous and accessible monitoring of cardiovascular vital signs – heart rate (HR), heart rate variability (HRV), and blood pressure (BP) – is a key enabler of early diagnosis, risk stratification, and long-term management of CVDs. Wrist-worn wearables based on optical photoplethysmography (PPG) have made such continuous monitoring widely accessible; however, optical PPG suffers from several well-documented limitations that reduce its reliability in everyday, real-world use [5–8]:

- **Motion artifacts:** physical movement and instability of the wearable device corrupt the acquired optical signal.
- **Subcutaneous fat thickness:** light scattering in thicker adipose layers reduces penetration depth and signal quality.
- **Skin pigmentation variability:** melanin absorption alters signal amplitude across different skin tones, introducing systematic bias.
- **Ambient light interference:** external light sources and associated noise affect signal quality.
- **Sweat and skin-sensor interface degradation:** moisture interrupts light transmission, particularly during physical activity.

These limitations motivate the exploration of alternative, non-contact sensing modalities – in particular, millimeter-wave (mmWave) radar – that are less sensitive to skin optical properties and ambient lighting, and that can potentially provide more robust arterial pulse sensing across diverse users and activity conditions.

1.2 Problem Statement

Radar-based sensing offers an emerging, non-contact solution for physiological monitoring. Recent systems such as AirBP and BP3 have demonstrated that mmWave radar can

be used to estimate blood pressure from wrist- or desk-mounted radar sensors [1,2]. However, a review of the existing literature reveals two key limitations in current radar-based physiological monitoring approaches:

1. Existing radar-based physiological studies are predominantly conducted in *static environments*, with subjects seated or lying still, which does not reflect real-world wearable use during daily activity.
2. Conventional radar systems rely on *fixed processing pipelines* and static sensing parameters, despite the fact that signal quality varies significantly with subject motion, range to the artery, and environmental interference.
3. There is no *direct, controlled comparison between optical PPG and mmWave radar arterial pulse sensing* under conditions where optical PPG is known to fail – such as during motion, in subjects with darker skin pigmentation, or with greater subcutaneous fat thickness – making it unclear whether radar-based sensing actually offers a robustness advantage over optical PPG in precisely the scenarios where it is most needed.

There is therefore a gap in the literature for a wearable radar-based sensing system that can dynamically adapt its sensing and processing parameters in response to real-time signal quality, while jointly supporting multiple downstream physiological estimation tasks (HR, HRV, and BP) within a single, generalized platform.

1.3 Objectives

The proposed project aims to achieve the following objectives:

- Develop a wearable adaptive radar based Arterial pulse sensing system prototype with closed-loop control to dynamically adjust sensing and processing parameters based on real-time signal quality.
- Develop a unified framework for joint heart rate, heart rate variability, and blood pressure estimation within the same sensing system.
- Demonstrate robust physiological monitoring across:
 - diverse activity conditions – sitting, walking, and running;
 - varying skin pigmentation;
 - varying subcutaneous fat thickness.

1.4 Scope

The scope of this project covers the following:

1. Design and development of a multimodal sensing module prototype, combining an Frequency Modulated Continuous Wave (FMCW) radar, an optical PPG front-end, and an IMU on a single wrist-worn platform.

2. Implementation of an adaptive radar controller that performs closed-loop optimization of sensing parameters under dynamic motion and interference conditions.
3. Prototype-based data collection from human volunteers under controlled conditions.
4. Development of an FMCW radar signal-processing pipeline for arterial pulse waveform extraction.
5. Validation of the derived physiological estimates against clinical-grade reference devices:
 - Polar H10 chest strap [9] – heart rate and heart rate variability ground truth;
 - Omron wrist-cuff blood pressure monitor [10] – blood pressure ground truth.
6. Evaluation of system performance across variations in skin tone, body composition, and activity level.

1.4.1 Novelty and the Uniqueness of the Project

The novelty of this project lies in addressing two specific gaps left open by the existing radar-based physiological monitoring literature:

1. **Adaptive mmWave sensing:** Existing radar-based physiological monitoring studies generally use fixed radar sensing and processing parameters. To the best of our knowledge, no prior work has developed a *closed-loop adaptive radar controller* that dynamically re-tunes sensing parameters in real time based on signal quality and motion state.
2. **mmWave sensing under dynamic conditions:** Existing radar-based physiological monitoring studies have been conducted primarily under *static or controlled conditions*, with limited validation during real-world activities. The robustness of mmWave-based arterial pulse sensing during *dynamic activities*, such as walking and running, remains insufficiently investigated, as most existing studies have focused on static or controlled conditions.
3. **Direct optical PPG vs. mmWave radar comparison under PPG-failure conditions:** To the best of our knowledge, there is no existing study that directly and systematically compares optical PPG against mmWave radar arterial pulse sensing under the specific conditions in which optical PPG is known to degrade – namely motion artifacts, darker skin pigmentation, and greater subcutaneous fat thickness. Consequently, it remains unclear from the literature which modality performs better precisely where optical PPG fails. This project aims to fill that gap through a controlled, head-to-head evaluation of both modalities on the same wearable platform and the same subjects.

1.4.2 Beneficiaries and Potential Applications

The outcomes of this project are expected to benefit a range of stakeholders and enable several downstream applications:

1. **Individuals with cardiovascular risk factors:** More accurate, motion-robust, and skin-tone-independent continuous cardiovascular monitoring can support earlier detection of abnormal heart rate, heart rate variability, and blood pressure patterns, enabling timely clinical intervention.
2. **Wearable device manufacturers:** Findings on the relative robustness of mmWave radar versus optical PPG can inform the design of next-generation wearables that combine or select sensing modalities based on activity context and user skin tone, improving accuracy for a broader and more diverse user population.
3. **Researchers:** The adaptive radar control framework and the optical PPG–mmWave radar comparison dataset generated by this project can serve as a foundation for future research on downstream tasks such as sleep apnea detection and stress monitoring using the same sensing platform.

Chapter 2

Literature Review

This chapter reviews prior work relevant to the proposed system, spanning wearable mmWave-radar-based arterial pulse sensing, cuffless blood pressure estimation, and adaptive radar waveform selection. A comparative summary against the proposed approach is presented at the end of the chapter.

2.1 Wearable Non-Invasive Arterial Pulse Detection with a Millimeter-Wave Radar

Bahmani et al. [3] developed a wrist-worn mmWave radar prototype and a signal-processing pipeline that extracts arterial pulse waveforms from raw intermediate frequency (IF) radar data, in parallel with a green-LED optical PPG channel and a continuous non-invasive arterial pressure (CNAP) reference. Figure 2.1a shows the experimental setup reported in this work, in which the wrist-worn radar prototype is worn alongside the optical PPG and CNAP reference sensors to enable a synchronized, direct morphology comparison between these two modalities; this is *not* the system model proposed in this project, but the reference experimental configuration from [3] that motivates and informs our own prototype design.

Their signal-processing pipeline, shown in Figure 2.1b, performs mean removal, range FFT, arctangent demodulation, phase unwrapping, band-pass filtering, peak detection, and antenna bin selection to recover the radar-derived inter-beat interval (IBI) and pulse waveform, in parallel with simpler peak-detection-based pipelines for the PPG and CNAP channels. A key finding of this work, illustrated by the arterial waveform morphology analysis in Figure 2.1c, is that wrist-worn mmWave radar can reconstruct arterial pulse waveforms with morphological characteristics that more closely resemble the clinical gold-standard CNAP arterial waveform than optical PPG, across different subjects. This result directly motivates the use of radar as a complementary, and potentially superior, sensing modality to optical PPG for waveform-based physiological monitoring.

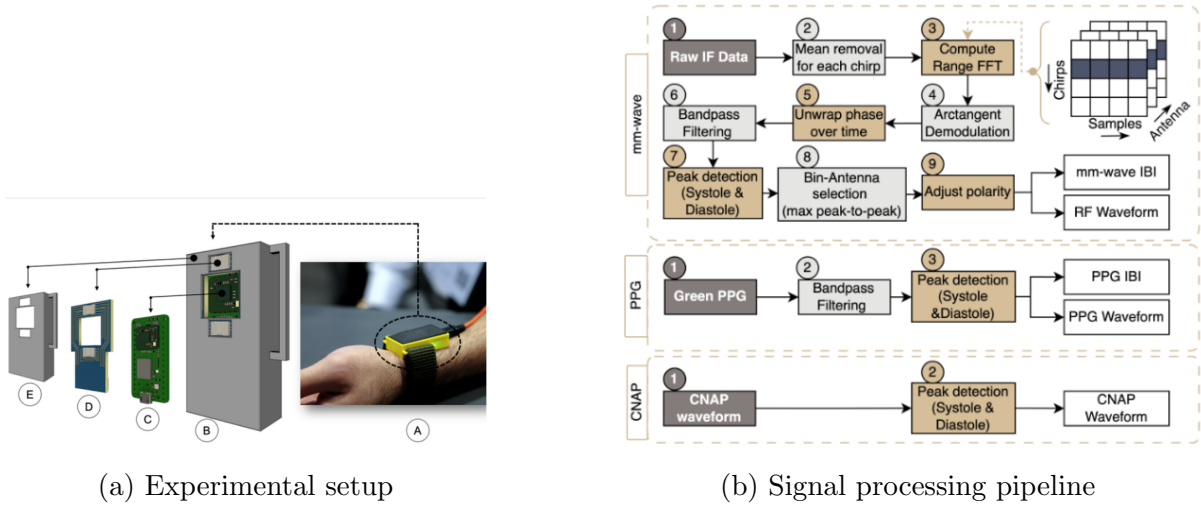
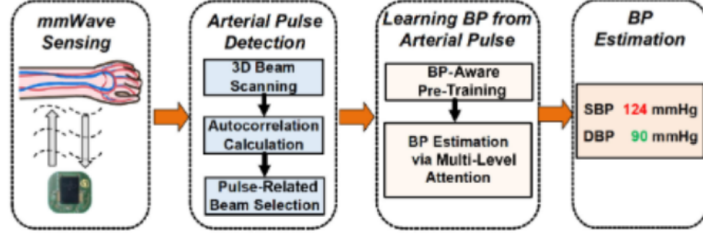


Figure 2.1: Overview of the system and arterial waveform analysis for comparing optical PPG and mmWave sensing, including the experimental setup, signal processing pipeline, and arterial waveform morphology analysis [3].

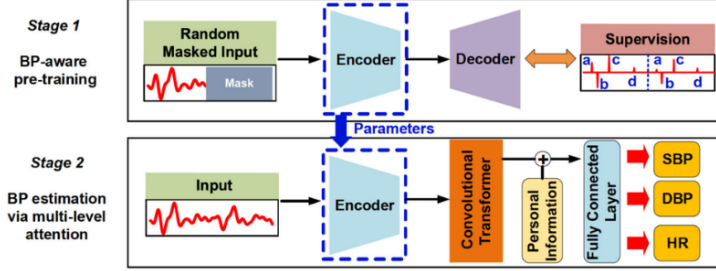
2.2 airBP: Monitor Your Blood Pressure with Millimeter-Wave in the Air

Liang et al. [2] proposed airBP, a contactless blood pressure monitoring system based on 3D beam-scanning mmWave sensing. Figure 2.2a shows the system overview reported in this work. As shown in the figure, the system performs autocorrelation-based computation to identify the radial-artery-related beam and range bin, followed by reflected-signal-strength-based pulse waveform extraction. The extracted pulse waveform is then fed into the two-stage machine learning pipeline shown in Figure 2.2b: a masking-based pre-training stage, in which fiducial points on the pulse waveform supervise a encoder-decoder to learn features important for BP estimation, followed by a second stage that uses the learned encoder’s latent representation together with a convolutional transformer and personal physiological information (e.g., age, height, weight, BMI) to estimate subject-independent systolic and diastolic blood pressure. AirBP’s key contribution is the development of a BP estimation framework using autocorrelation-based radial artery range-bin selection and reflected-signal-strength-based pulse waveform extraction, though evalua-

tion is restricted to static desk-mounted scenarios.



(a) System overview

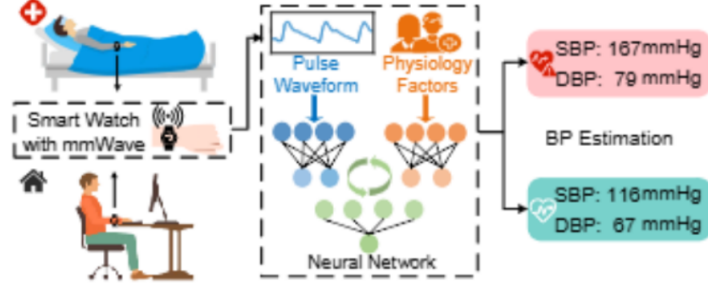


(b) ML pipeline

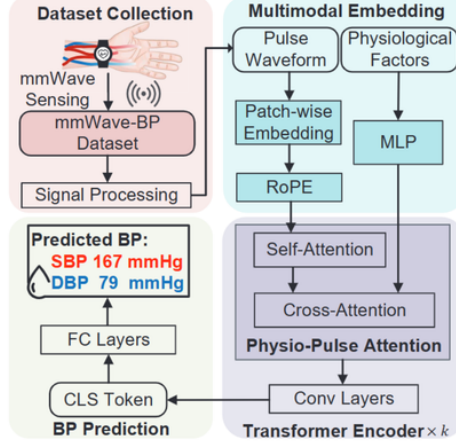
Figure 2.2: AirBP system overview and machine learning pipeline for cuffless blood pressure estimation [2].

2.3 BP3: Improving Cuff-less Blood Pressure Monitoring Performance by Fusing mmWave Pulse Wave Sensing and Physiological Factors

Zheng et al. [1] presented BP3, a smartwatch-integrated mmWave radar system for cuffless blood pressure (BP) monitoring. Figure 2.3a shows the system overview reported in this work, in which mmWave-derived pulse-waveform features are combined with physiological metadata, including gender, age, height, weight, BMI, and BP history, to estimate BP. The corresponding machine learning pipeline is shown in Figure 2.3b: the pulse waveform is represented using patch-wise embeddings with rotary positional encoding (RoPE), followed by self-attention, cross-attention with the physiological-factor embedding, and a physio-pulse attention transformer encoder for BP regression. By incorporating subject-level physiological context, BP3 improves cuffless BP estimation, including performance across subjects with abnormal BP and different BMI ranges. However, the primary contribution is centered on the deep-learning architecture and multimodal feature fusion, rather than on novel radar signal-processing or adaptive sensing techniques.



(a) System overview



(b) ML pipeline

Figure 2.3: BP3 System Overview and Machine Learning Pipeline for Wrist-Worn Cuffless Blood Pressure Estimation [1].

2.4 Constrained Contextual Bandit Learning for Adaptive Radar Waveform Selection

Thornton et al. [4] studied adaptive radar waveform selection in a general (non-physiological) radar sensing context, using a constrained contextual bandit formulation in which the transmitted waveform w_{t+1} is selected based on spectrum observations $\hat{s}_t$, waveform constraints $\hat{d}$, and receiver-side performance feedback y_t from the previous transmission w_t . Figure 3.5 shows the adaptive model proposed in this work: the transmitted waveform w_t propagates through the target channel (transmit channel, target, and receive channel), and the resulting receiver output y_t is fed back, together with a spectrum observation $\hat{s}_t$ and a waveform constraint $\hat{d}$, into a constrained contextual bandit waveform-selection block, which outputs the next waveform w_{t+1} to be transmitted, closing the loop between sensing outcome and transmit configuration on a per-transmission basis. The key finding of this work is that radar waveforms selected *dynamically* based on environmental observations and receiver-side performance feedback outperform radar systems using static waveform settings. This result forms the conceptual basis for the closed-loop, contextual-bandit-driven adaptive radar controller proposed in this project, adapted here to the specific reward signal of cardiac signal quality rather than general target-detection performance.

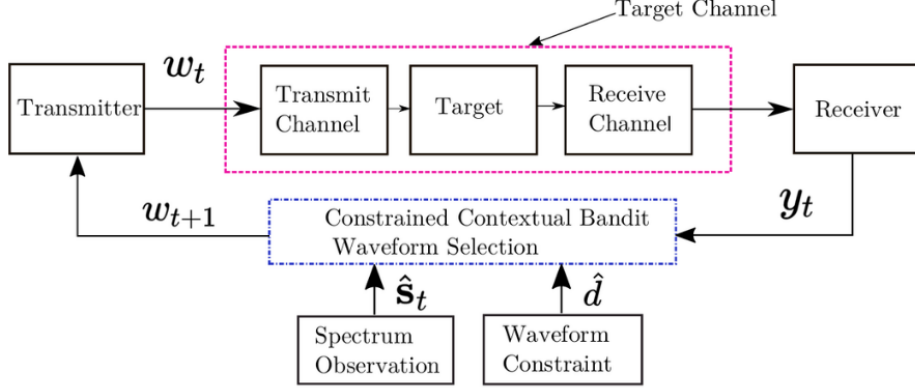


Figure 2.4: Adaptive Model [4].

2.5 Literature Comparison with the Proposed Approach

Table 2.1 summarises the reviewed radar-based physiological monitoring literature and compares it against the approach proposed in this project across five dimensions: sensing multimodality, presence of an adaptive radar controller, evaluation under static versus dynamic (walking/running) conditions, the downstream physiological tasks addressed, and the evaluation criteria used.

Table 2.1: Comparison of the proposed approach with existing radar-based physiological monitoring literature.

Approach	Multi-modality	Adaptive Radar	Condition	Downstream Tasks / Evaluation
Bahmani et al. [3]	mmWave & optical PPG	No	Static	HR: mmWave–PPG comparison
AirBP [2]	mmWave	No	Static	BP: BMI, skin tone and lighting variations
BP3 [1]	mmWave	No	Static	BP: abnormal BP and BMI
Ours	mmWave, optical PPG & IMU	Yes	Dynamic	HR, HRV, BP: <i>evaluated under motion, skin-tone and BMI variations</i>

As shown in Table 2.1, no existing approach simultaneously combines multimodal sensing (radar, optical PPG and IMU), an adaptive radar controller, evaluation under dynamic activity conditions, and joint HR/HRV/BP estimation with cross-modality robustness evaluation. This gap defines the novelty and contribution of the proposed project.

Chapter 3

Methodology

This chapter presents the methodology adopted to realise the proposed wrist-worn multi-modal sensing system. It begins with the overall system architecture and the mathematical model that relates the transmitted radar waveform to arterial wall motion, including a comparison of the two candidate approaches for recovering the pulse waveform from the received signal. It then covers the prototype architecture and the rationale behind the selection of each sensing component, the adaptive radar controller, and the signal-processing and machine learning pipeline. It concludes with the experimental setup, resource requirements, budget, task delegation, timeline, and deliverables that define how the project will be executed.

3.1 System Overview

The proposed system consists of a closed-loop sensing and processing chain, illustrated in Figure 3.1. A transmitter radiates an mmWave signal that reflects off the skin and underlying arterial tissue and is captured by the receiver. The received signal is passed through a signal-processing block to extract the arterial pulse waveform, which is fed into a lightweight machine learning (ML) block to jointly estimate heart rate/heart rate variability (HR/HRV) and diastolic/systolic blood pressure (DBP/SBP). In parallel, an adaptive algorithm block continuously evaluates receiver-side signal quality and issues a new radar configuration to the transmitter, closing the sensing loop.

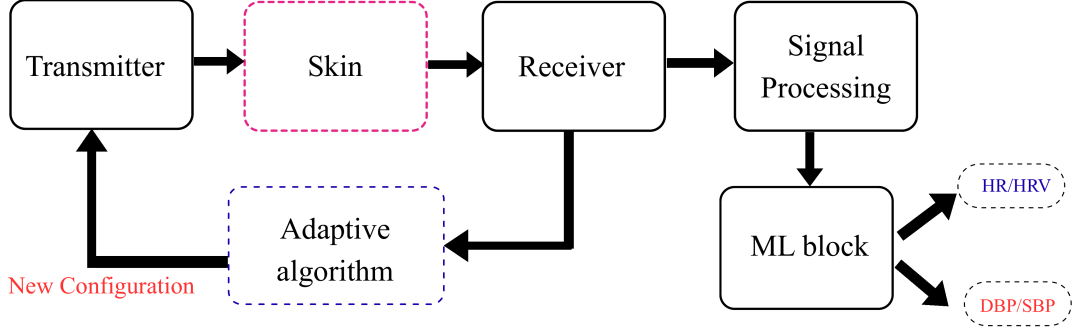


Figure 3.1: High-level system overview: closed-loop radar sensing, signal processing, and ML-based physiological estimation.

3.2 Mathematical Sensing Model

This section formalises the relationship between the transmitted frequency-modulated continuous-wave (FMCW) waveform and the arterial wall displacement that the system aims to recover. It then derives the two candidate methods for extracting the pulse waveform from the complex range-FFT output, amplitude-based and phase-based extraction, and establishes the basis on which the phase-based method was selected for this project.

3.2.1 Transmitted Signal

The radar transmits a linear FMCW chirp. Over one chirp of duration T_c , the instantaneous frequency increases linearly from the start frequency f_0 at a slope $S = B/T_c$, where B is the swept bandwidth:

$$s_T(t) = A_T \cos \left[2\pi \left(f_0 t + \frac{S}{2} t^2 \right) \right], \quad 0 \leq t \leq T_c. \quad (3.1)$$

For the BGT60TR13C, $f_0 = 58$ GHz and $B = 5.5$ GHz, giving a carrier wavelength $\lambda = c/f_0 \approx 5$ mm at the band centre.

3.2.2 Target Model

The illuminated target is the skin surface above the radial artery. Its range is modelled as a static standoff d_0 (sensor-to-skin distance) superimposed with a small time-varying displacement $\Delta d(t)$ caused by arterial wall pulsation, and a motion-induced term $\Delta d_m(t)$ arising from wrist movement:

$$d(t) = d_0 + \Delta d(t) + \Delta d_m(t). \quad (3.2)$$

The arterial component $\Delta d(t)$ is quasi-periodic at the cardiac fundamental frequency f_c and is of the order of tens of micrometres, whereas $\Delta d_m(t)$ is broadband and may be several orders of magnitude larger. Separating the two is the central sensing challenge addressed by the adaptive controller of Section 3.4.

3.2.3 Received Signal and Beat Frequency

The reflected signal arrives delayed by the round-trip propagation time $\tau(t) = 2d(t)/c$ and attenuated by a factor determined by path loss and target reflectivity:

$$s_R(t) = A_R \cos \left[2\pi \left(f_0(t - \tau) + \frac{S}{2}(t - \tau)^2 \right) \right]. \quad (3.3)$$

Mixing $s_R(t)$ with $s_T(t)$ and low-pass filtering yields the intermediate-frequency (IF), or *beat*, signal:

$$s_B(t) = A_B \cos(2\pi f_b t + \phi), \quad f_b = \frac{2Sd}{c}, \quad \phi = \frac{4\pi d}{\lambda}. \quad (3.4)$$

The beat frequency f_b encodes target range, while the beat phase ϕ encodes range at sub-wavelength precision. A range fast Fourier transform (FFT) across the fast-time samples of each chirp separates targets into range bins of width

$$\Delta R = \frac{c}{2B} \approx 2.7 \text{ cm} \quad (3.5)$$

for $B = 5.5 \text{ GHz}$. The range-FFT output for bin k at slow-time index m is a complex quantity,

$$X[k, m] = I[k, m] + jQ[k, m], \quad (3.6)$$

from which two distinct observables may be formed: the magnitude $|X| = \sqrt{I^2 + Q^2}$ and the phase $\phi = \tan^{-1}(Q/I)$. The choice between them defines the two extraction approaches compared in the following subsections.

3.2.4 Approach A: Amplitude-Based Extraction

The first approach discards the phase and forms the pulse waveform from the magnitude of the selected range bin:

$$x_{\text{RSS}}[m] = |X[k_{\text{wrist}}, m]|. \quad (3.7)$$

At first inspection this appears unable to sense arterial motion at all, since Equation 3.4 places displacement information in the phase term. The sensitivity arises indirectly through coherent summation. The receiver does not observe the arterial reflection in isolation; the bin contains the vector sum of a large static component C , arising from skin, tissue, nearby surfaces, and radar package coupling, and a much smaller arterial return of amplitude A and phase ϕ :

$$X = C + Ae^{j\phi}, \quad A \ll |C|. \quad (3.8)$$

Taking C real and positive without loss of generality, the squared magnitude is

$$|X|^2 = C^2 + A^2 + 2CA \cos \phi, \quad (3.9)$$

which for $A \ll C$ reduces to $|X| \approx C + A \cos \phi$. Perturbing about an operating point $\phi_0 = 4\pi d_0/\lambda$ by a small arterial phase excursion $\Delta\phi$ gives

$$|X| \approx C + A \cos \phi_0 - \underbrace{A \sin \phi_0}_{\text{sensitivity}} \Delta\phi. \quad (3.10)$$

The amplitude waveform is therefore proportional to the arterial phase excursion, and hence to displacement, but scaled by the factor $-A \sin \phi_0$. Amplitude-based methods such as BP3 [1] exploit precisely this leakage.

3.2.5 Approach B: Phase-Based Extraction

The second approach retains the complex bin value and tracks its phase directly. Differentiating the phase term of Equation 3.4 with respect to range gives

$$\Delta\phi = \frac{4\pi \Delta d}{\lambda}. \quad (3.11)$$

At $\lambda = 5$ mm, an arterial wall displacement of $\Delta d = 25 \mu\text{m}$ produces a phase change of approximately 3.6° , which is well above the phase noise floor of the transceiver. Tracking ϕ over slow time across successive chirps recovers the arterial pulse waveform:

$$x_\phi[m] = \text{unwrap}\{\angle X[k_{\text{wrist}}, m]\}, \quad (3.12)$$

where k_{wrist} is the bin selected by the range-selection stage of the pipeline. Phase unwrapping is required because Equation 3.11 wraps for displacements exceeding $\lambda/4 \approx 1.25$ mm between consecutive samples; arterial pulsation remains well inside this limit, although gross wrist motion does not.

Range resolution and displacement sensitivity are governed by different mechanisms and are therefore independent quantities. Although $\Delta R \approx 2.7$ cm is three orders of magnitude coarser than arterial wall motion, it constrains only the ability to *separate* two targets in range, not the ability to *track* the motion of a single target within one bin.

3.3 Prototype Architecture

The wrist prototype integrates three sensing modalities an FMCW radar, an optical PPG front end, and an IMU around a central microcontroller unit (MCU), as shown in Figure 3.2. Acquired data is streamed from the MCU to a host laptop over a WiFi module for offline processing.

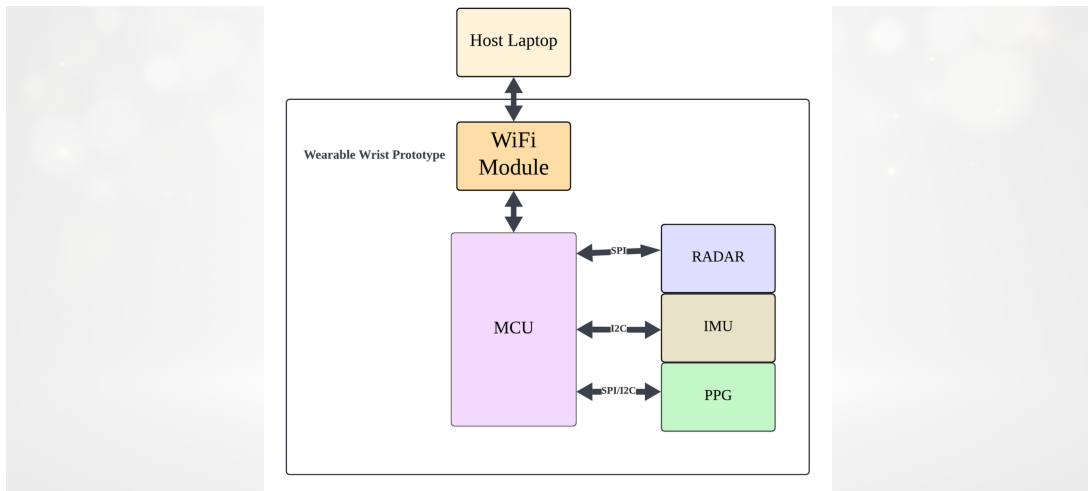


Figure 3.2: Prototype architecture of the wrist device.

3.3.1 Radar Transceiver Selection

Three candidate mmWave radar transceivers were evaluated for the prototype: IWR1443 and IWR6843, and Infineon’s BGT60TR13C. Table 3.1 compares the three candidates on bandwidth, unit and evaluation-board cost, and antenna configuration, while Table 3.2 compares raw ADC access, package integration, and physical footprint.

Table 3.1: Comparison of candidate mmWave radar transceivers

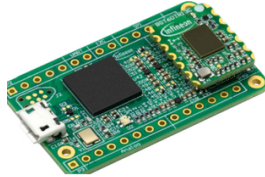
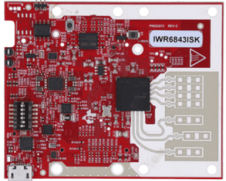
Radar	IWR1443 [11]	BGT60TR13C [12]	IWR6843 [13]
Radar Image			
Bandwidth	76–81 GHz	58.5–63.5 GHz	60–64 GHz
IC Price	\$32.96	\$19.72	\$32.59
Evaluation Board Price	\$483.48	\$224.65	\$220.50
Antenna Configuration	3 TX 4 RX	1 TX 3 RX	3 TX 4 RX

Table 3.2: Integration and form-factor comparison of candidate radar transceivers

Attribute	IWR1443	BGT60TR13C	IWR6843
Raw ADC access	Yes (LVDS, via DCA1000)	Yes (SPI, on-chip FIFO)	Yes (LVDS, via DCA1000)
Antenna-in-package	No	Yes	No
Evaluation board dimensions	80.8 × 65.2 mm	64.0 × 25.4 mm	134.6 × 70.1 mm

The BGT60TR13C was selected as the radar transceiver for this project based on its cost, integration, and form-factor advantages. It offers the lowest IC and evaluation-board cost among the considered candidates and integrates the antenna within the package, eliminating the need for high-frequency PCB routing at 60 GHz and thereby reducing associated design and manufacturing risks. Furthermore, its compact board footprint of 64 mm × 25.4 mm is significantly smaller than the alternatives, making it the only candidate considered practically suitable for integration into a wrist-worn form factor.

3.3.2 Optical PPG Sensor Selection

Two candidate optical PPG front end ICs were evaluated, as summarized in Table 3.3.

Table 3.3: Comparison of candidate optical PPG sensors

Sensor	MAX86176	MAX30101
LED Support	Up to 6 (configurable)	Fixed 3 (IR/Green/Red)
Photodiode Inputs	Up to 4	Fixed 1
ADC Resolution	20-bit	18-bit
Waveform Fidelity	High (clinical)	Moderate (consumer)

The MAX86176 was selected for its high ADC resolution, configurable LED and photodiode channels, and clinical-grade waveform fidelity. These features enable reliable acquisition of high-quality PPG signals, providing a suitable reference for a like-for-like comparison with the pulse waveform derived from the radar system.

3.3.3 Radar Operating Parameters

Following the configuration reported in BP3 [1], the initial radar operating parameters for the BGT60TR13C are set as summarized in Table 3.4. These parameters form the default (non-adaptive) baseline configuration, around which the adaptive radar controller (Section 3.4) dynamically adjusts sensing parameters.

Table 3.4: Baseline FMCW radar operating parameters. The range resolution follows from Equation 3.5 and does not limit displacement sensitivity, which is governed instead by Equation 3.11.

Parameter	Value
Bandwidth	58–63.5 GHz
ADC Sample Rate	2 MHz
Samples per Chirp	32 (minimum)
Pulse Signal Sample Rate	200 Hz
Range Resolution	2.7 cm

3.3.4 Optical PPG Wavelength Selection

Both red (660 nm) and infrared (940 nm) wavelengths carry information related to blood oxygenation, since deoxygenated hemoglobin (Hb) absorption exceeds oxygenated hemoglobin (HbO₂) absorption at 660 nm, while the reverse relationship holds at 940 nm. Green light (530 nm) is strongly absorbed by hemoglobin and is therefore the primary channel used for heart rate monitoring and for direct comparison against the radar-derived pulse waveform [14].

3.4 Adaptive Radar Controller

The central novelty of the proposed system is a closed-loop adaptive radar controller that re-tunes the radar’s sensing parameters every coherent processing interval (CPI), driven by a signal-quality reward and the wearer’s current motion state, as illustrated in Figure 3.3.

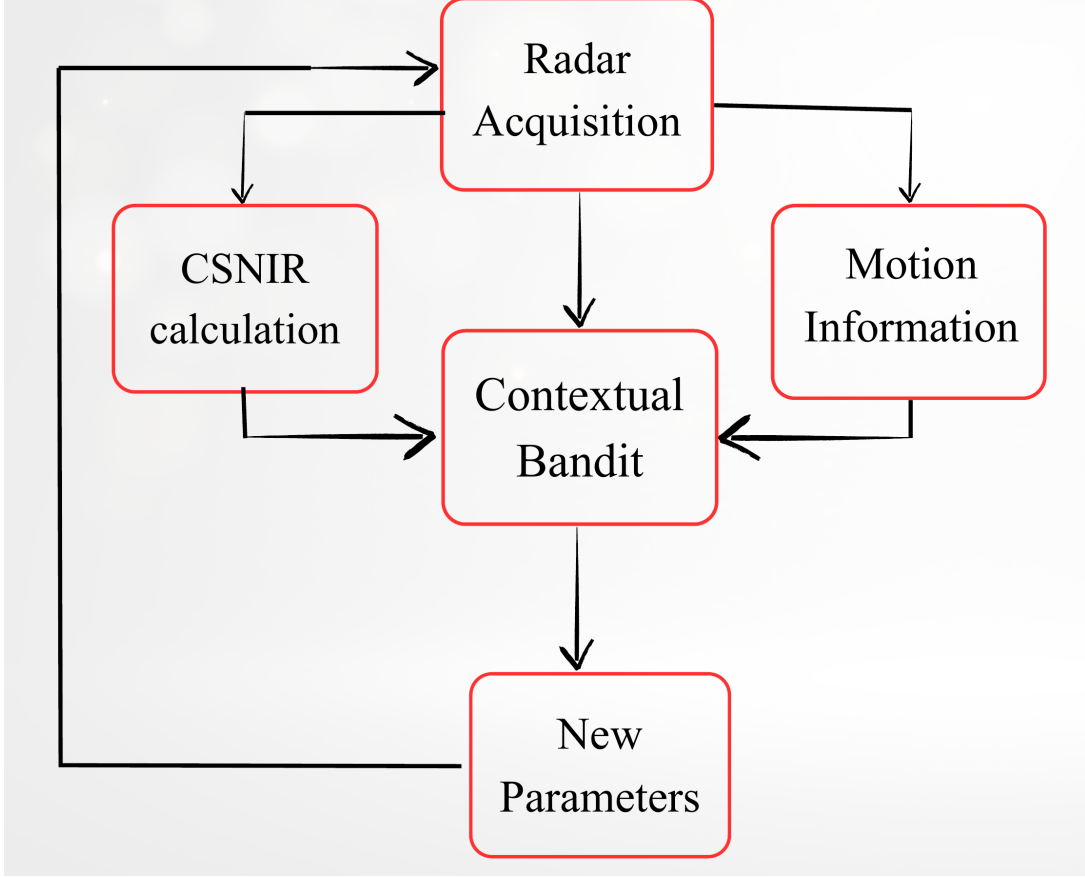


Figure 3.3: Closed-loop adaptive radar controller.

Following each radar acquisition, two quantities are computed: (i) the cardiac signal-to-interference-plus-noise ratio (CSINR) of the acquired data, and (ii) the current motion state, derived from the IMU. Both are supplied as context to a contextual bandit algorithm [4], which selects a new radar configuration (e.g., chirp rate, bandwidth, or gain settings) to maximise expected CSINR under the current motion condition. The selected configuration is applied to the radar for the next CPI, closing the loop. This design directly addresses the key limitation identified in the literature review: that conventional radar-based physiological monitoring systems rely on fixed processing pipelines despite significant motion-induced variation in signal quality.

The controller is specified by three elements:

- **Reward metric:** the cardiac signal-to-interference-plus-noise ratio (CSINR), defined in Equation 3.13.
- **Reconfiguration cadence:** per coherent processing interval (CPI).
- **Selection algorithm:** contextual bandit.

$$\text{CSINR} = 10 \log_{10} \left(\frac{\int_{f_c - \Delta f}^{f_c + \Delta f} S_x(f) df}{\int_{f \notin [f_c - \Delta f, f_c + \Delta f]} S_x(f) df} \right) \quad (3.13)$$

where:

$S_x(f)$ Power spectral density of the phase-derived pulse waveform $x_\phi[m]$ of Equation 3.12, extracted from the selected range bin and computed over one CPI.

f_c Cardiac fundamental frequency for that CPI, estimated from the dominant peak of the autocorrelation function $R_{xx}(\tau)$.

Δf Half-bandwidth of the band treated as signal around f_c (starting value ≈ 0.1 Hz, covering the expected beat-to-beat drift within a CPI).

The numerator is the spectral power concentrated in the narrow band around the cardiac fundamental, and the denominator is the spectral power everywhere else in the CPI, which combines motion artefact and noise without requiring the two to be separated. The ratio is expressed in decibels via $10 \log_{10}(\cdot)$, consistent with the standard SINR convention.

3.5 Signal Processing and Machine Learning Pipeline

The radar-side signal-processing and ML pipeline, shown in Figure 3.4, transforms raw IF ADC data into a cleaned pulse waveform suitable for physiological estimation, implementing Approach B of Section 3.2.5.

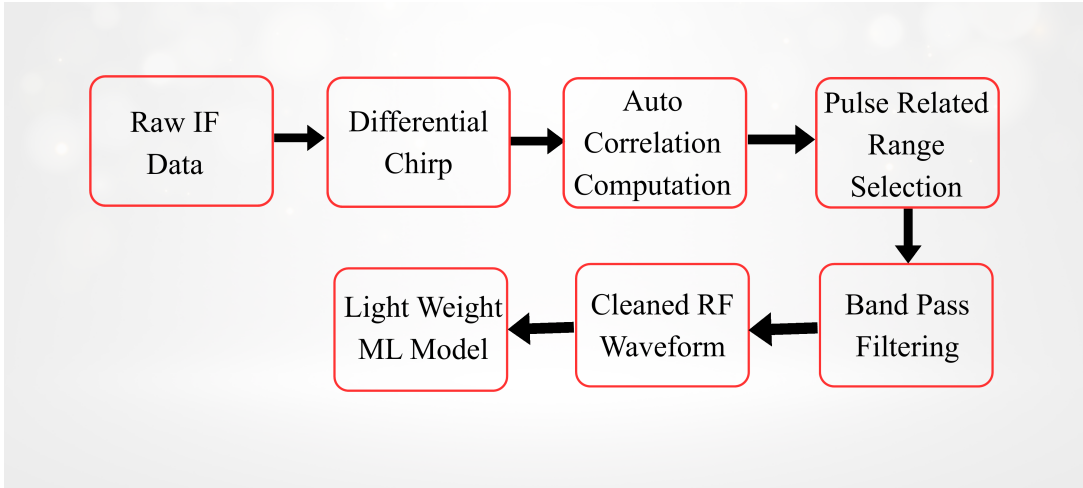


Figure 3.4: Radar signal-processing and machine learning pipeline.

The pipeline proceeds through the following stages:

1. **Raw IF data** is acquired per chirp across all receive antennas, as described by Equation 3.4.
2. **Range FFT** is applied across the fast-time samples of each chirp, producing the complex range profile $X[k, m]$ of Equation 3.6. The complex output is retained rather than reduced to magnitude, preserving the phase term required by Equation 3.11.
3. **Differential chirp** processing is applied to suppress static clutter and isolate time-varying arterial motion.

4. **Autocorrelation computation** is performed to identify range bins exhibiting strong periodic (cardiac) structure.
5. **Pulse-related range selection** identifies the range bin k_{wrist} corresponding to the radial artery, based on autocorrelation peak strength, following the approach used in AirBP [2].
6. **Phase extraction and unwrapping** recovers the displacement-proportional waveform $x_\phi[m]$ of Equation 3.12 from the selected bin.
7. **Band-pass filtering** restricts the signal to the physiologically relevant cardiac frequency band, removing residual noise, baseline wander, and interference.
8. The result is a **cleaned RF pulse waveform**, structurally analogous to an optical PPG waveform, representing the arterial blood-volume-pulse signal.
9. Finally, a **lightweight ML model** is applied to the cleaned waveform to jointly estimate HR, HRV, and BP.

For the optical PPG and IMU channels, parallel lightweight processing pipelines are used: the PPG signal is band-pass filtered and subjected to peak detection to identify systolic and diastolic fiducial points and derive inter-beat intervals, while the IMU signals are processed to characterise motion-related activity and artefacts.

3.6 Experimental Setup and Reference Standards

Figure 3.5 shows the planned experimental setup for data collection and validation. Each subject wears three devices simultaneously: a Polar H10 chest strap (ECG reference, used to derive HR/HRV ground truth), an Omron blood-pressure cuff on the left wrist (BP reference), and the custom mmWave radar/PPG/IMU prototype on the right wrist.



Figure 3.5: Planned experimental setup showing reference devices and the custom prototype.

- **Polar H10 chest strap** [9] – provides continuous single-lead ECG, used as the ground-truth reference for HR and HRV.

- **Omron wrist-cuff blood pressure monitor (BP4350)** [10] – an FDA-approved oscillometric device, used as the ground-truth reference for systolic and diastolic blood pressure.



(a) Polar H10 chest strap



(b) Omron BP cuff

Figure 3.6: Reference devices used for physiological measurements.

3.7 Uniqueness of the Proposed Approach

The proposed project differentiates itself from existing work in the following ways:

1. It incorporates an *adaptive radar controller* that dynamically re-configures sensing parameters, rather than relying on static parameters across all conditions.
2. Existing mmWave physiological monitoring studies have primarily focused on static or controlled conditions. In contrast, this work *evaluates mmWave sensing under both static and dynamic conditions, including walking and running*.
3. It conducts a systematic mmWave radar–optical PPG comparison across *diverse skin pigmentation, subcutaneous fat thickness, and activity conditions* to directly evaluate the relative sensing robustness of the two modalities.
4. It combines *three sensing modalities* (radar, optical PPG, and IMU) in a single wrist-worn platform, whereas prior work addresses at most two.

3.8 Resource Requirements

3.8.1 Hardware

- **Prototype components:** BGT60TR13C mmWave radar, MAX86176 optical PPG sensor, and an IMU for motion tracking.
- **Reference standards:** Polar H10 chest strap (ECG ground truth) and Omron BP4350 wrist-cuff monitor (reference BP).

3.8.2 People and Facilities

- 30 healthy volunteers for prototype-based data collection.
- Ethics clearance from the University of Moratuwa for human-subject data collection.

3.9 Estimated Budget

Table 3.5 presents the estimated hardware budget for the project.

Table 3.5: Estimated project budget

Component	Unit	Price (USD)
Industrial mmWave Radar Sensor	4	\$78.88
Evaluation Board for the Radar Sensor	1	\$248.92
PPG Sensor	3	\$43.71
IMU Sensor	3	\$12.69
MCU	3	\$60.00
PCB, components, and enclosure	—	\$200.00
Total		\$644.20

3.10 Task Delegation

Table 3.6 outlines the delegation of project tasks among the four group members. Each member holds primary responsibility for a distinct technical subsystem, while data collection and documentation are shared across the group.

Table 3.6: Task delegation among group members.

Task	P1	P2	P3	P4
Signal-processing pipeline	✓			
Machine-learning architecture	✓			
Adaptive algorithm design		✓		
Skin-tone & fat-thickness study		✓		
Sensor integration & synchronisation			✓	
Acquisition & wireless streaming			✓	
Component selection & power design				✓
PCB layout & enclosure				✓
Data collection & experiments	✓	✓	✓	✓
Validation, documentation & writing	✓	✓	✓	✓

P1: Ananthakumar T. **P2:** Sivamynthan N. **P3:** Mathujan S. **P4:** Kopithan M.

3.11 Project Timeline

The project is planned over an approximately eight-month period (September 2026 to April 2027), structured into six phases: ethics and approval, design and algorithm development, pilot validation using the radar evaluation kit, hardware iteration (two PCB revisions), full-scale data collection and model training, and final analysis and report writing, as shown in Figure 3.7. Key milestones include submission of the PCB v1 design for fabrication, a mid-project evaluation following hardware finalisation, and a final demonstration.

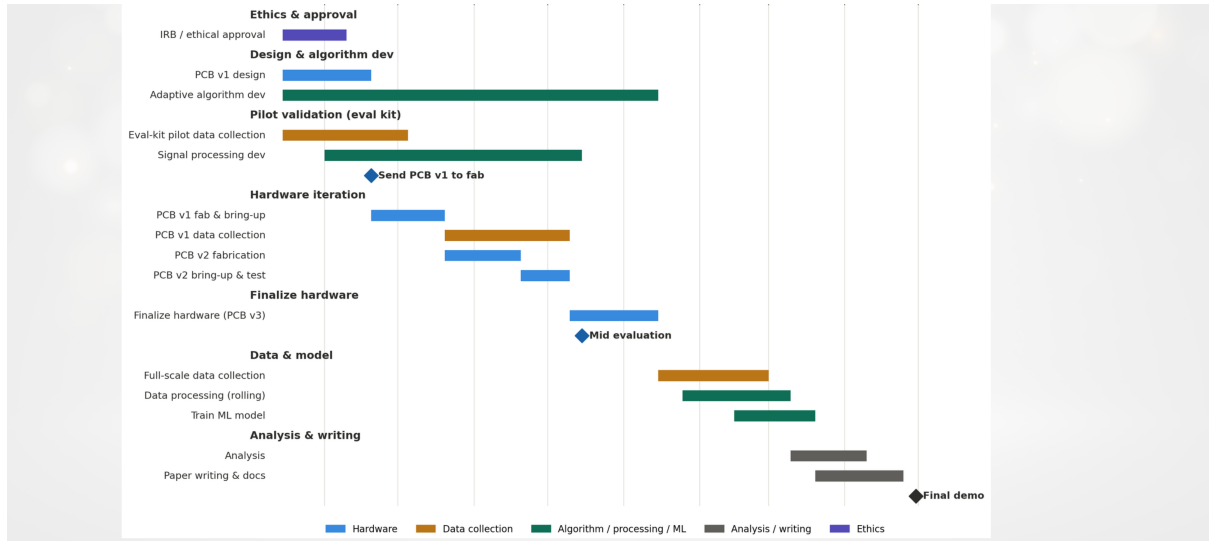


Figure 3.7: Project Gantt chart and timeline (September 2026 – April 2027).

3.12 Project Deliverables

The project will produce three categories of deliverable: a physical sensing prototype, the algorithms required to operate it and extract physiological parameters, and a quantitative validation of the resulting estimates against established reference instruments.

3.12.1 Hardware

- A functional multimodal sensing module prototype, integrating the mmWave radar, optical photoplethysmography (PPG), and inertial measurement unit (IMU) front-ends on a wrist-worn platform.

3.12.2 Algorithm

- An adaptive radar control framework.
- An offline signal-processing pipeline for heart rate (HR), heart rate variability (HRV), and staged blood pressure (BP) extraction.

3.12.3 Validation

- Accuracy benchmarks for HR and HRV against Polar H10 electrocardiogram (ECG) ground truth.
- Accuracy benchmarks for BP against the FDA-approved Omron BP4350 wrist-cuff reference.
- A performance comparison dataset evaluating radar versus PPG robustness across skin tone, body composition (fat layer), and activity level.

3.13 Preliminary Work

The following preliminary work has already been carried out to establish initial feasibility:

- The BGT60TR13C evaluation kit was tested in a wrist-worn configuration.
- Infineon’s Radar Fusion GUI was used to configure chirp/frame parameters and to record raw ADC data from the evaluation kit.
- Ethics application documents have been prepared for submission to the University of Moratuwa ethics review board.
- ECG data acquired from the Polar H10 chest strap was successfully visualised, confirming reliable acquisition of the HR/HRV reference signal (Figure 3.8).

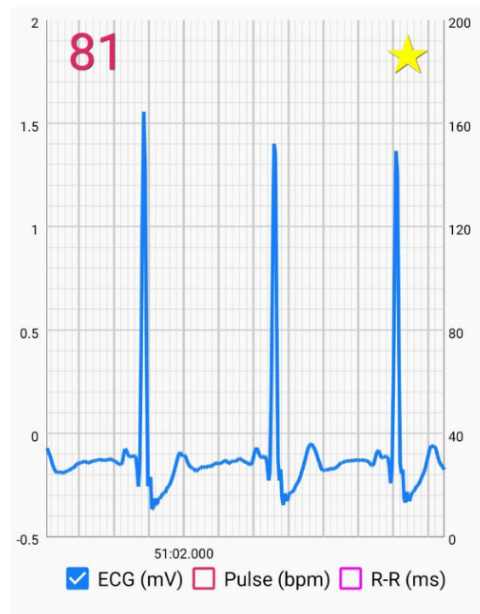


Figure 3.8: ECG waveform acquired from the Polar H10 chest strap during preliminary testing.

This chapter established the technical and organisational basis of the project. The FMCW sensing model was derived, and two candidate methods for recovering the arterial pulse waveform from the complex range-FFT output were compared: amplitude-based extraction, which senses displacement indirectly through coherent summation with static

clutter and carries a standoff-dependent gain that vanishes at multiples of $\lambda/4$, and phase-based extraction, which yields displacement in absolute units with constant sensitivity. The latter was adopted. The BGT60TR13C radar and MAX86176 PPG front end were selected on the grounds of cost, package integration, and form factor; the adaptive radar controller and the phase-based signal-processing pipeline were defined as the core technical contributions; and the experimental protocol, budget, task allocation, and timeline were set out to demonstrate that the proposed work is achievable within the available resources and schedule.

Chapter 4

Discussion

4.1 Technical Feasibility

The technical feasibility of the proposed system rests on three main pillars: (i) the availability of a commercially accessible, wearable-form-factor mmWave radar transceiver, (ii) the demonstrated ability of prior work to extract arterial pulse waveforms and estimate BP from radar-derived signals, and (iii) the tractability of implementing a closed-loop contextual-bandit controller on embedded hardware.

The selected radar transceiver, the BGT60TR13C, has an antenna-in-package design and a board footprint of only $64\text{ mm} \times 25.4\text{ mm}$, making it realistically suited to a wrist-worn enclosure – a constraint that ruled out the larger IWR1443 and IWR16843 evaluation boards. Notably, the same wrist-worn sensing configuration has already been used in existing peer-reviewed research [3], confirming that reliable arterial pulse sensing is achievable with this exact hardware class rather than being a purely speculative design choice. Preliminary testing using the BGT60TR13C evaluation kit in a wrist-worn configuration, together with successful acquisition and visualization of Polar H10 ECG data, provides early empirical evidence that the core sensing chain is achievable within the project timeframe. The signal-processing approach (differential chirp processing, autocorrelation-based range-bin selection, and band-pass filtering) draws directly on peer-reviewed methods already validated in AirBP [2] and Bahmani et al. [3], reducing algorithmic risk. In addition, the selected radar transceiver exposes fully *configurable* chirp, bandwidth, and gain parameters via SPI at run time, which is a necessary hardware precondition for implementing the proposed closed-loop adaptive radar controller. The main remaining technical uncertainty lies in the adaptive controller itself: while contextual bandit algorithms have been shown to outperform static radar configurations in a general radar sensing context [4], their application to a cardiac-SINR reward signal on resource-constrained wearable hardware, updated at CPI rate, has not been previously demonstrated and will require careful latency and stability analysis during implementation.

4.2 Financial Feasibility

The financial feasibility of the project is supported by both cost and procurement considerations. The estimated hardware budget of \$644.20 is modest and dominated by the radar evaluation board and PCB fabrication costs. The Polar H10 chest strap, used as the

HR/HRV reference device, is already supported and provided by the external supervisor, removing this item from the procurement budget entirely. The mmWave radar sensor, the highest-lead-time component, has already been purchased and is currently in transit, which substantially de-risks the project schedule with respect to import delays. The remaining components – the PPG sensor, IMU, MCU, and PCB/enclosure materials – are low-cost, commercially available from standard distributors, and can be procured and assembled within the available project budget and timeframe without requiring additional external funding.

4.3 Social Feasibility

The proposed project is socially feasible on several grounds. First, data collection involves only non-invasive, external sensing (radar, optical PPG, IMU, ECG chest strap, and cuff-based BP measurement), posing no more risk to volunteers than routine consumer wearable or clinical vital-sign measurement, and will proceed only after ethics clearance and informed consent are obtained from all participants. Second, recruiting 30 healthy volunteers from the university community is realistic given the accessible, quick, and non-invasive nature of each measurement session. Third, the project directly addresses a socially relevant equity gap: existing optical PPG wearables are known to be less accurate for users with darker skin tones and higher body fat, and a validated radar-based alternative has the potential to make wearable cardiovascular monitoring more accurate and inclusive across diverse user populations. Finally, given the growing global burden of cardiovascular disease, a low-cost, wearable, activity-robust monitoring solution is well aligned with broader public health goals of enabling earlier detection and more equitable access to continuous cardiovascular monitoring.

4.4 Ethical Considerations

As the project involves collecting physiological data (radar reflections, PPG signals, IMU data, ECG, and blood pressure readings) from human volunteers, ethics clearance from the University of Moratuwa is a mandatory prerequisite for data collection and has already been initiated during the preliminary work phase. All data collection will follow informed consent procedures, and collected data will be anonymized and used solely for the stated research objectives.

4.5 Risk Factors and Mitigation

Table 4.1 summarizes the key risks identified for this project, together with their corresponding mitigation strategies.

Table 4.1: Project risk factors and mitigation strategies

Risk	Mitigation
Limited pigmentation and body composition diversity among available volunteers.	Measure and report skin tone within the available volunteer range, and interpret results accordingly.
Custom PCB bring-up fails or requires multiple design spins.	Use of an antennas-in-package radar IC to reduce 60 GHz routing risk; two PCB iterations are planned into the schedule.
Component lead times and import delays.	Prioritize the first radar sensor purchase early to allow time for international shipping and customs delays.
Delay in ethics approval may affect the overall project timeline.	Prepare all required ethics documentation and the experimental protocol well in advance of data collection.

4.6 Impact of the Project

4.6.1 Local Impact

Within Sri Lanka, cardiovascular disease remains a leading cause of morbidity and mortality, yet continuous, activity-robust vital-sign monitoring remains largely inaccessible outside urban clinical settings. A locally developed wearable sensing platform capable of reliable operation during daily activity could help extend continuous cardiovascular monitoring beyond hospital and clinic settings, supporting earlier detection of abnormal cardiovascular patterns for individuals who do not have regular access to specialist care. The project also strengthens local research capacity in mmWave radar sensing and embedded physiological signal processing at the University of Moratuwa, contributing reusable hardware designs, signal-processing pipelines, and a validated radar-PPG comparison dataset that can seed further student and faculty research in wearable health technology. Additionally, by explicitly evaluating performance across the skin pigmentation range represented in the local volunteer population, the project generates evidence directly relevant to the accuracy of wearable devices for Sri Lankan and South Asian users, a demographic that is often underrepresented in the datasets used to develop and validate commercial optical PPG wearables.

4.6.2 Global Impact

More broadly, cardiovascular disease is the leading cause of death worldwide, responsible for nearly 20 million deaths in 2022 alone. Wearable optical PPG devices, while widely deployed globally, are known to suffer from systematic accuracy degradation for users with darker skin pigmentation, higher subcutaneous fat thickness, and during motion – limitations that disproportionately affect specific demographic groups and active users. By providing direct, controlled evidence on whether and when mmWave radar sensing outperforms optical PPG under exactly these failure conditions, this project contributes toward closing a well-documented equity and reliability gap in global wearable health technology. The proposed closed-loop adaptive radar controller and unified HR/HRV/BP

estimation framework also represent a generalizable contribution to the wearable sensing research community, offering a validated methodology that can be adopted or extended by future radar-based physiological monitoring systems worldwide. In the longer term, more accurate and inclusive continuous cardiovascular monitoring can support earlier detection of cardiovascular abnormalities at a population scale, aligning with global public health priorities around reducing the worldwide burden of cardiovascular disease.

Chapter 5

Conclusion

This report has presented the feasibility study for a wearable, multimodal mmWave radar and optical PPG sensing system for robust, non-contact arterial pulse monitoring. Motivated by the global burden of cardiovascular disease and the well-documented limitations of optical PPG sensing – motion artifacts, ambient light interference, sweat, skin pigmentation variability, and subcutaneous fat thickness – this project proposes to develop a wrist-worn prototype that integrates an FMCW mmWave radar (BGT60TR13C), a clinical-grade optical PPG front-end (MAX86176), and an IMU, coupled with a closed-loop adaptive radar controller and a unified signal-processing and machine-learning pipeline for joint heart rate, heart rate variability, and blood pressure estimation.

A review of existing radar-based physiological monitoring literature (Bahmani et al. [3], AirBP [2], BP3 [1]) shows that, while radar-based sensing is a promising and validated alternative to optical PPG, existing systems are evaluated exclusively under static conditions and rely on fixed processing pipelines. Combined with evidence from the general radar sensing literature that adaptive, feedback-driven waveform selection outperforms static configurations [4], this motivates the central contribution of the proposed project: a closed-loop, contextual-bandit-driven adaptive radar controller evaluated under dynamic, real-world motion conditions.

The proposed system has been assessed as technically, financially, and operationally feasible within the scope of a final year project. Component selection has been finalized based on a systematic comparison of candidate radar transceivers and PPG sensors, an experimental validation protocol against clinical-grade reference devices (Polar H10 and Omron BP4350) has been defined, and preliminary testing with the radar evaluation kit has already demonstrated successful wrist-worn radar acquisition and reference ECG visualization. An estimated budget of \$644.20 and a eight-month project timeline (September 2026 – April 2027) have been established, along with identified risks and corresponding mitigation strategies.

Upon completion, this project is expected to deliver a functional multimodal sensing prototype, an adaptive radar control framework, an offline signal-processing pipeline for HR, HRV, and staged BP extraction, and a validated performance comparison of radar versus optical PPG sensing robustness across skin tone, body composition, and activity level – directly addressing the identified gap in current radar-based wearable physiological monitoring research.

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